Probaho — Architecture

Draft generated from the checked-in Topic E implementation; review with the team before submission.

Scope

A local-first, synthetic SME cash-flow recovery workflow: scoped records, deterministic projections, bounded plan comparison, approval, synthetic supplier consent, sandbox execution, read-back, and evidence.

Runtime path

Browser UI → loopback HTTP API → validated domain functions → scoped SQLite → deterministic projections and plan comparisons → snapshot-bound approval → supplier-consent gate → idempotent sandbox receipt.

Projection

Integer BDT paisa; base, late-collection, and approved-recovery scenarios; pre-opening history excluded; paid/collected invoices excluded; daily reserve and payroll checks; source references attached to events.

Agent boundary

Scoped requests expose read_ledger, project_cash, compare_plans, and create_reminder_draft only. The model cannot approve, execute, send money, call arbitrary APIs, or access raw record descriptions.

Evidence

Plan hash, snapshot hash, policy version, approval fields, consent fields, action fingerprints, before/after state, projection before/after, read-back, idempotency key, and per-business SHA-256 audit chain.

Limitations

No real bank, lender, supplier, multi-user authentication, encrypted storage, or production compliance. The synthetic sandbox demonstrates control flow, not deployment readiness.